Linear Buckling of a Cylindrical Stiffened Panel

Workflow, theory, and a structural-sizing exercise with the JFEM case-runner web app

JFEM

Universidad de Sevilla

Exercise 07.1 ▪ June 13, 2026

What you will do

Goals

- Run a **SOL 105 linear buckling** analysis of a curved, stringer-stiffened panel from the browser.
- Read the geometry, **loads**, **boundary conditions**, and **reference axes** correctly.
- Understand the theory behind the buckling eigenvalue λ and its limitations.
- **Size the structure**: change thickness or laminate so the lowest factor $\lambda_1 \approx 1.0$.

The tool

A pure-Julia HTTP server plus a single-page browser app: `APPS/PANDEATOR_APP/`.

It builds a Reference (BDF) deck, runs OpenJFEM in-process, and renders mode shapes, the eigenvalue list, and the total model mass in 3D.

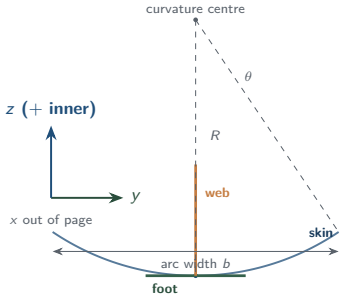
The physical model: a cylindrical stiffened panel

- A **curved skin**: an arc of radius R and developed width b, length L.
- A central **"T" stringer** running along x at the crown: a **web** (blade, height h) standing on the inner side, and a flat **foot** (flange, width w) bonded to the crown.
- Three shell property regions: skin, web, foot.
- Meshed with `CQUAD4` shells; the web base and foot centreline *share* the skin crown nodes, so the parts act as one structure.

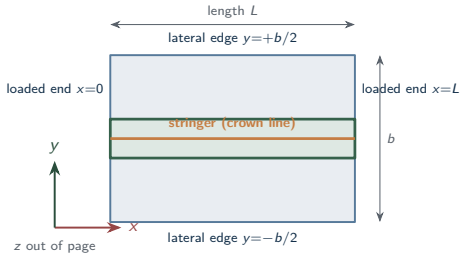
Why this case?

It is the canonical aerospace "one bay of a stiffened shell": compression and shear can trigger *local skin buckling*, *stringer crippling*, or *global panel buckling* – exactly what a sizing engineer must keep above the applied load.

Reference axes and geometry



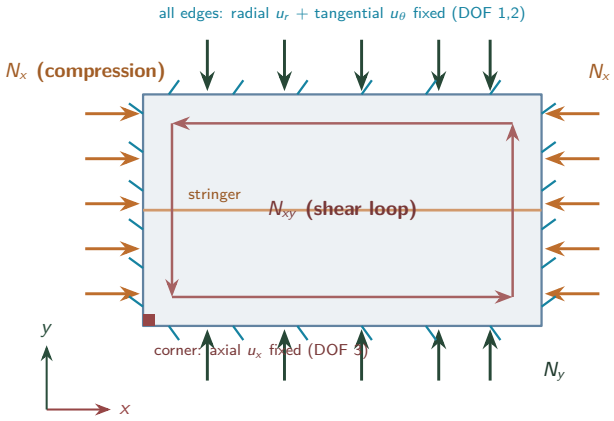
(a) cross-section (y–z)



(b) plan (x–y)

Basic rectangular frame: x = length (stringer axis), y = arc-width direction, z = radial (curvature centre at $z = R$; crown at $z = 0$; $+z$ points to the inner/concave side). Each `GRID` also references a cylindrical `CORD2C` frame (`CD=1`) so edge constraints and loads act in a **radial / tangential / axial** system.

Loads and boundary conditions



Force flows are entered in **N/mm** and lumped to edge nodes in the cylindrical frame $[\,r,\ \theta,\ x\,]$. All are **self-equilibrated** so no net reaction is needed.

Loads and boundary conditions – the details

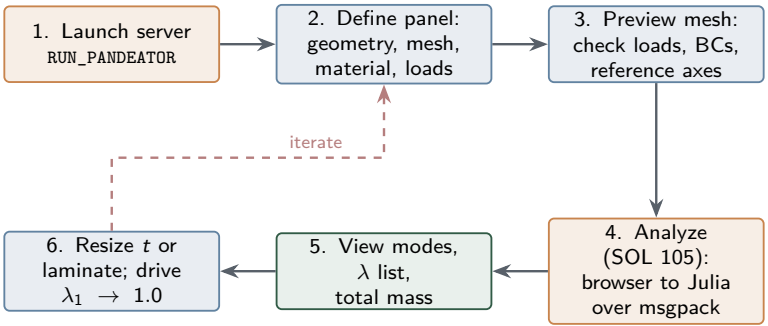
Applied loads (force flow)

- N_x – axial compression: equal and opposite on the two ends, pushing inward; applied to skin + web + foot end nodes.
- N_{xy} – shear flow: a circulating, head-to-tail loop around the perimeter (closed $\Rightarrow$ zero net force/moment).
- N_y – transverse compression: inward tangential on the two lateral edges.

Constraints (assembly-like)

- **All four edges**: radial u_r + tangential u_θ fixed (DOF 1,2). The panel cannot bulge out-of-plane or fan out hoop-wise – the neighbouring bays hold it.
- Axial (x) is left **free** on the edges so the panel can shorten under compression.
- **One corner**: a single axial (u_x, DOF 3) point lock removes the last rigid-body freedom. The load is self-equilibrated, so this lock carries ≈ 0 reaction.

Workflow of the web app



1 start the launcher (`RUN_PANDEATOR_WINDOWS.cmd` / `RUN_PANDEATOR_MAC_LINUX.sh`); **2** fill in the sidebar form; **3** *Preview mesh* to verify the model; **4** press the floating *Analyze* button (top-right, shown when inputs change); **5** inspect modes, the λ list, and total mass; **6** iterate the sizing.

Running it – step by step

1. **Launch the server.** Double-click `RUN_PANDEATOR_WINDOWS.cmd` (or run `./RUN_PANDEATOR_MAC_LINUX.sh`). The browser opens at `http://127.0.0.1:8088/`.
2. **Choose "Build stiffened panel"** (the default analysis source).
3. **Set geometry/mesh/material/loads** in the sidebar. Defaults give a reasonable aluminium panel.
4. **Preview mesh** – check the load arrows, BC cones, and the reference axes all read as expected.
5. **Analyze (SOL 105)** – the deck is sent to Julia; a modal shows the live solve time.
6. **Read the results** – the *Result* box lists each buckling mode; the **eigenvalue list** below it shows the factors λ_k; the **total mass** appears in the lower-right corner.

What λ means

The applied force flows are multiplied by λ to reach the critical (buckling) load. $\lambda_1 < 1$ **means the panel buckles below the applied load** – unsafe. $\lambda_1 = 1$ is the margin-zero sizing point.

Theory: linear (eigenvalue) buckling

The two-step SOL 105 procedure

1. **Static preload.** Solve $\mathbf{K}\,\mathbf{u}_0 = \mathbf{f}$ for the reference load $\mathbf{f}$ (the applied force flows). This gives the membrane stress state in every element.
2. **Differential stiffness.** From that stress state, assemble the **geometric (differential) stiffness** $\mathbf{K}_g(\mathbf{u}_0)$, which captures how membrane forces stiffen or soften out-of-plane bending.
3. **Eigenproblem.** Find the load multipliers λ and mode shapes ϕ that make the tangent stiffness singular.

$$\left(\mathbf{K} + \lambda\,\mathbf{K}_g\right)\phi \;=\; \mathbf{0}, \qquad \mathbf{K}\,\phi \;=\; -\,\lambda\,\mathbf{K}_g\,\phi.$$

The smallest positive λ_1 is the **critical load factor**; ϕ_1 is the buckling shape. In the deck this is `SOL 105` with a static `SUBCASE`, `STATSUB` linking the preload, and `EIGRL` selecting the lowest modes.

Theory: why it is an eigenvalue problem

- Total potential energy = strain energy − work of loads. Equilibrium is a stationary point; **stability** is whether that point is a minimum.
- The tangent stiffness is $\mathbf{K}_T(\lambda) = \mathbf{K} + \lambda\,\mathbf{K}_g$. As the load grows ($\lambda \uparrow$), membrane compression makes $\mathbf{K}_g$ subtract from $\mathbf{K}$.
- Buckling is the load at which $\mathbf{K}_T$ first loses positive-definiteness, i.e. $\det(\mathbf{K} + \lambda\,\mathbf{K}_g) = 0$.
- $\mathbf{K}_g$ is *linear in the preload stress*, so the critical loads scale as eigenvalues λ of the applied pattern.

Reading the number

- λ_1 = factor on *all* applied flows simultaneously.
- Buckling **load** = $\lambda_1 \times$ (applied N_x, N_{xy}, N_y).
- Higher modes $\lambda_2, \lambda_3, \ldots$ are further bifurcations; the first one governs.

Theory: a hand check (intuition)

For a simply-supported *flat* rectangular plate of thickness t under uniaxial compression, the classical critical stress is

$$\sigma_{cr} \;=\; k\,\frac{\pi^2 E}{12\,(1-\nu^2)}\left(\frac{t}{b}\right)^2,$$

so the critical *force flow* $N_{cr} = \sigma_{cr}\,t \propto t^3$.

This t^3 scaling is the lever you will use in the exercise: a small thickness change moves λ a lot.

Curvature & stiffeners help

- Curvature (R) raises the critical load above the flat-plate value.
- The stringer subdivides the skin and adds bending stiffness, raising λ further and changing the mode shape.
- The web can itself buckle (crippling) – watch which part moves in ϕ_1.

Exercise: size the panel to $\lambda_1 \approx 1.0$

Task

Starting from the default panel and loads, **adjust the structure** until the lowest buckling factor is on the order of **1.00** (target band $0.95 \le \lambda_1 \le 1.05$), then report the resulting **mass**.

Isotropic track (MAT1). Vary the *skin thickness* t_{skin} (and, if needed, web/foot thickness).

- Recall $N_{cr} \propto t^3$: if $\lambda_1 = 2.7$ at $t = 2.0\,\text{mm}$, then $t_{\text{new}} \approx t\,(1/\lambda_1)^{1/3}$ is a good first guess.
- Re-run, read the new λ_1, and refine once or twice.

Composite track (MAT8/PCOMP). Keep the load fixed and change the *laminate*.

- Add/remove plies, e.g. `(45/-45/0/90)S` → `(45/-45/0/90)2S`.
- Re-orient plies toward the load direction to raise λ for the same ply count.
- Note that mass scales with total ply count.

Exercise: a suggested procedure

1. **Baseline.** Use the defaults (R=1000, b=600, L=800 mm; skin/web/foot t=2.0 mm; aluminium; N_x=200 N/mm). Analyze and record λ_1 and the total mass.
2. **Predict.** If $\lambda_1 > 1$ the panel is over-strong (you can lighten it); if $\lambda_1 < 1$ it is under-strong (you must add material). Use the t^3 rule for a first thickness estimate.
3. **Iterate.** Change *one* parameter, re-Analyze, read the new λ_1. Two or three iterations should land inside the target band.
4. **Report.** Record final parameters, λ_1, the governing mode shape (skin / web / global), and the **mass** – the headline sizing trade.
5. **Reflect.** Which change bought the most λ per gram? How did the mode shape move?

Deliverable

A one-page table: baseline vs. final (parameters, λ_1, mass), plus a screenshot of the governing buckling mode.

Exercise: worked example of the sizing logic

Lightening an over-strong panel

Suppose the baseline gives $\lambda_1 = 2.70$ at $t_{\text{skin}} = 2.0\,\text{mm}$. The panel is $2.7\times$ stronger than needed.

$$t_{\text{target}} \approx t \left(\frac{1}{\lambda_1}\right)^{1/3} = 2.0 \left(\frac{1}{2.70}\right)^{1/3} \approx 1.43 \text{ mm}.$$

Set $t_{\text{skin}} \approx 1.4\,\text{mm}$, re-Analyze, then nudge by $\pm 0.05\,\text{mm}$ until λ_1 is in $[0.95, 1.05]$.

Caveats for the estimate

- The t^3 rule is exact only for an isolated flat plate; curvature and the stringer shift it – so always confirm by re-running.
- If the mode shape is *web* buckling, thinning the skin will not fix λ; thicken the web instead.
- Composite: thickness is set by ply count, so changes are discrete.

Limitations of linear (SOL 105) buckling

What it assumes

- **Linear, elastic** material; small pre-buckling deflections.
- Buckling load $= \lambda \times$ a *fixed* reference stress pattern (proportional loading).
- **Perfect geometry** – no initial imperfections, no eccentricities.
- The bifurcation is detected from the *undeformed* preload state.

Why the real load is lower

- **Imperfection sensitivity**: curved/shell structures buckle well below the linear λ (knock-down factors).
- **Material nonlinearity**: yielding caps the usable load before elastic buckling.
- **Post-buckling & large deflection**: ignored here (use SOL 106 / nonlinear).
- **Mesh & BC idealisation**: results depend on mesh density and the assumed edge restraint.

Limitations: practical guidance

- Treat λ_1 as an **upper bound** on the true buckling strength. For a real margin, apply a knock-down factor and check yielding separately.
- **Refine the mesh** until λ_1 stops changing (mesh convergence) before trusting the number.
- Inspect the **mode shape**: local skin, web crippling, and global panel modes need different fixes.
- Default boundary conditions model *one bay of a larger assembly*. If your panel is a free edge instead, the restraints – and λ – change.
- For final sizing, validate the worst cases against a reference solver and, where post-buckling matters, a nonlinear analysis.

Method provenance

The element formulations and solution-sequence choices used by OpenJFEM are documented in the companion review *"JFEM Method Origins and Bibliography"* (in `Reference_documentation/`).

Summary

You learned to

- Drive the case-runner web app end to end for SOL 105.
- Interpret the reference axes, loads, and BCs of the curved stiffened panel.
- Read the buckling factor λ and total mass.
- Size thickness / laminate to a target $\lambda_1 \approx 1.0$.
- Recognise the limitations of linear buckling.

Key takeaway

λ_1 is the factor between the applied load and elastic buckling. Sizing to $\lambda_1 \approx 1$ means *margin-zero*; in practice keep a reserve and account for imperfections and yield.

JFEM Exercise 07.1 ▪ `APPS/PANDEATOR_APP` ▪ SOL 105 linear buckling